

Lesson 10: Numerical Methods for Solving Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 10

Video: Numerical Methods for Solving Differential Equations **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that in the real world of engineering and aerospace, most nonlinear differential equations have no closed-form analytical formula. When exact integration is impossible, numerical algorithms march step-by-step through the slope field. Euler's method uses a simple tangent line approximation; Heun's method averages the starting and predicted ending slopes; and the industry gold-standard RK4 computes a weighted average of four trial slopes to achieve remarkable fourth-order accuracy.

2. Core Theoretical Framework & Rigorous Formulations

1. Forward Euler's Method

$$y_{n+1} = y_n + hf(x_n, y_n), \quad x_{n+1} = x_n + h$$

Local truncation error: $\mathcal{O}(h^2)$; Global cumulative error: $\mathcal{O}(h)$.

2. Improved Euler (Heun's Predictor-Corrector Method)

- Predictor: $y_{n+1}^* = y_n + hf(x_n, y_n)$
- Corrector: $y_{n+1} = y_n + \frac{h}{2} [f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*)]$

3. Runge-Kutta 4th Order (RK4)

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = f(x_n, y_n)$, $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$, $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$, and $k_4 = f(x_n + h, y_n + hk_3)$. Global error: $\mathcal{O}(h^4)$.

3. Fully Solved Whiteboard Problems

Exam Drill: Comparative Stepping with Euler vs Improved Euler

Problem Statement: Given $y' = x + y$ with $y(0) = 1$, compute $y(0.2)$ using a step size $h = 0.1$ via (a) Forward Euler and (b) Improved Euler (Heun). Compare against exact analytical solution $y(x) = 2e^x - x - 1$.

Step 1: Forward Euler Step 1 ($x_0 = 0 \rightarrow x_1 = 0.1$)

$$f(0, 1) = 0 + 1 = 1. \quad y_1 = 1 + 0.1(1) = 1.1000.$$

Step 2: Forward Euler Step 2 ($x_1 = 0.1 \rightarrow x_2 = 0.2$)

$$f(0.1, 1.1) = 0.1 + 1.1 = 1.2. \quad y_2 = 1.1 + 0.1(1.2) = 1.2200.$$

Step 3: Improved Euler Step 1 ($x_0 = 0 \rightarrow x_1 = 0.1$)

$$\text{Predictor: } y_1^* = 1.1. \quad \text{Corrector: } y_1 = 1 + \frac{0.1}{2}[f(0, 1) + f(0.1, 1.1)] = 1 + 0.05[1 + 1.2] = 1 + 0.05(2.2) = 1.1100.$$

Step 4: Improved Euler Step 2 ($x_1 = 0.1 \rightarrow x_2 = 0.2$)

$$f(0.1, 1.11) = 1.21. \quad \text{Predictor: } y_2^* = 1.11 + 0.1(1.21) = 1.231. \quad \text{Corrector: } y_2 = 1.11 + 0.05[1.21 + (0.2 + 1.231)] = 1.11 + 0.05[2.641] = 1.2421.$$

Step 5: Exact Comparison

$$\text{Exact value: } y(0.2) = 2e^{0.2} - 0.2 - 1 \approx 2(1.22140) - 1.2 = 1.2428. \quad \text{Euler error: } |1.2428 - 1.2200| = 0.0228; \quad \text{Heun error: } |1.2428 - 1.2421| = 0.0007 \text{ (over 30x more accurate!)}.$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never evaluate the corrector step using the previous x_n coordinate. Heun's corrector requires $f(x_{n+1}, y_{n+1}^*)$ evaluated at the advanced coordinate $x_{n+1} = x_n + h$.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

Euler is 1st-order; Heun is 2nd-order; RK4 is 4th-order. Halving h in RK4 slashes global error by a factor of 16 (2^4).

- Video reference: [Numerical Methods for Solving Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.